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9203/1

Paper 1

Mark scheme

June 2022

Version: 1.0 Final Mark Scheme



2 2 6 Y 9 2 0 3 / 1 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from [oxfordaqaexams.org.uk](https://www.oxfordaqaexams.org.uk)

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Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make their judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which do not form the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent (for example, a scientifically correct answer that could not reasonably be expected from a student's knowledge of the specification).

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**.
Alternative words in the mark scheme are shown by a solidus eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates a correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name **two** magnetic materials.

[2 marks]

Student	Response	Marks awarded
1	iron, steel, tin	1
2	cobalt, nickel, nail*	2

3.2 Use of symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, or uses symbols to denote quantities in a physics equation, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. At any point in a calculation students may omit steps from their working. If a subsequent step is given correctly, the relevant marks may be awarded.

Full marks are **not** awarded for a correct final answer from incorrect working.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

An error can be carried forward from one question part to the next and is shown by the abbreviation 'ecf'.

Within an individual question part, an incorrect value in one step of a calculation does not prevent all of the subsequent marks being awarded.

3.6 Phonetic spelling

Marks should be awarded if spelling is not correct but the intention is clear, **unless** there is a possible confusion with another technical term.

3.7 Brackets

(.....) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do **not** accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

3.11 Numbered answer lines

Numbered lines on the question paper are intended to support the student to give the correct number of responses. The answer should still be marked as a whole.

4. Level of response marking instructions

Extended response questions are marked on level of response mark schemes.

- Level of response mark schemes are broken down into levels, each of which has a descriptor.
- The descriptor for the level shows the average performance for the level.
- There are two or three marks in each level.

Before you apply the mark scheme to a student's answer; read through the answer and, if necessary, annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1: Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level.

The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer. Do **not** look to penalise small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level.

Use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2: Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do **not** have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

You should ignore any irrelevant points made. However, full marks can be awarded only if there are no incorrect statements that contradict a correct response.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question 1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	55		1	AO1 3.7.1 f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	78		1	AO1 3.7.1 f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3	electrons are negative and protons are positive	allow electrons and protons have opposite charge	1	AO1 3.7.1 c d
	there are the same number of protons and electrons		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4	protons		1	AO1 3.7.1 f
	neutrons		1	
		this order only		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5	300 000 000 = 9 190 000 000 × λ		1	AO2 3.3.1 h
	$\lambda = \frac{300\,000\,000}{9\,190\,000\,000}$	allow a correct rearrangement of an incorrect power of 10 for distance or frequency	1	
	λ = 0.032644 (m)	allow a correct calculation using an incorrect power of 10 for distance or frequency	1	
	λ = 0.0326 (m)	allow their calculated value of λ rounded to 3 sig figs an answer of 0.03 (m) with correct working scores 3 marks	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.6	1 second		1	AO2 3.7.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.7	resolution has increased	allow a description of how the resolution has changed ignore references to accuracy or precision	1	AO3 3.7.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.8	improvement in technology or so time can be measured/known more accurately		1	AO3 3.7.1

Total Question 1		13
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Question 2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	A – cornea B – lens C – retina	allow 1 mark for 1 or 2 correct	2	AO1 3.3.6 h

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	large pupils allow more light into the eye and owl monkeys need to see in the dark	allow pupils control the amount of light entering the eye	1 1	AO1 3.3.6 h

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	long sight	allow hyperopia allow hypermetropia	1	AO1 3.3.6 j

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	the lens is too close to the retina		1	AO3 3.3.6 j

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.5	human eyes only detect visible light	allow human eyes cannot detect both infrared radiation and ultraviolet radiation	1	AO1 3.3.6 h

Question	Answers	Mark	AO / Spec. Ref.
02.6	Level 2: Scientifically relevant features are identified; the way(s) in which they are similar / different is made clear and (where appropriate) the magnitude of the similarity / difference is noted.	3–4	AO1 3.3.6 I
	Level 1: Relevant features are identified and differences noted.	1–2	
	No relevant content	0	
	Indicative content Similarities • they both have lens • both produce a real image • both produce an inverted image Differences • the CCD sensor / film in the camera acts like the retina in the eye • the aperture in the camera acts like the pupil in the eye • to focus the image on the retina the shape of the eye lens is changed • the lens of the camera can be changed • to focus the image on the CCD sensor / film the distance of the lens in the camera is changed • the eye has a blind spot but a CCD does not to access level 2 there must be a similarity and a difference identified.		
Total Question 2		11	

Question 3

Question	Answers	Mark	AO / Spec. Ref
03.1	Level 3: The design / plan would lead to the production of a valid outcome. All key steps are identified and logically sequenced.	5–6	AO1 3.5.2
	Level 2: The design / plan would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.	3–4	
	Level 1: The design / plan would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	
	No relevant content	0	
	Indicative content Constructing the electromagnet • coil the wire around the iron bar • connect the ends of the wire to the power supply using the leads with crocodile clips • use a low potential difference to limit current and avoid overheating, but sufficiently large to attract paper clips Carrying out the experiment • record the number of turns in the coil • turn the power supply on • use the electromagnet to pick up paper clips • count the number of paper clips picked up or measure the maximum distance at which a paperclip can be attracted • turn off the power supply • repeat experiment using different numbers of turns on the coil • repeat the whole experiment • keep the potential difference on the power supply the same each time Analysis • calculate the mean number of paper clips • plot a graph of number of turns against number of paperclips picked up • the more paperclips picked up the stronger the electromagnet to access level 3, students must describe how to make the electromagnet, vary the number of turns, and indicate how to measure the strength of the electromagnet.		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	5		1	AO2 3.5.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	a smaller change in the strength of the electromagnet could be detected		1	AO4 3.5.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	the more turns on the electromagnet the stronger the electromagnet	allow the more turns on the electromagnet the more paperclips attracted do not accept directly proportional	1	AO3 3.5.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	increase the potential difference (of the power supply)	allow voltage for potential difference	1	AO4 3.5.2

Total Question 3		10
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Question 4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	$t = 600$ seconds		1	AO2 3.1.2 c
	$2.6 = \frac{\text{distance}}{600}$	allow substitution of an incorrectly / not converted value of time	1	
	distance = 2.6×600	allow a correct rearrangement using an incorrectly / not converted value of time	1	
	distance = 1560 (m)	allow a correct calculation using an incorrectly / not converted value of time allow 1600 (m)	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	number of complete loops = $\frac{1560}{3.0}$	ecf from question 04.1	1	AO2 3.1.2
	number of complete loops = 520		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3	$0.10 = \frac{(3.0 - 2.6)}{t}$		1	AO2 3.1.3 d
	$t = \frac{0.4}{0.10}$		1	
	$t = 4$ (s)	allow 4.0 (s)	1	

Total Question 4		9
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Question 5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	an observed increase in wavelength	allow an observed decrease in frequency	1	AO1 3.8.3 a
	as an object moves away from us	allow as the Universe expands	1	

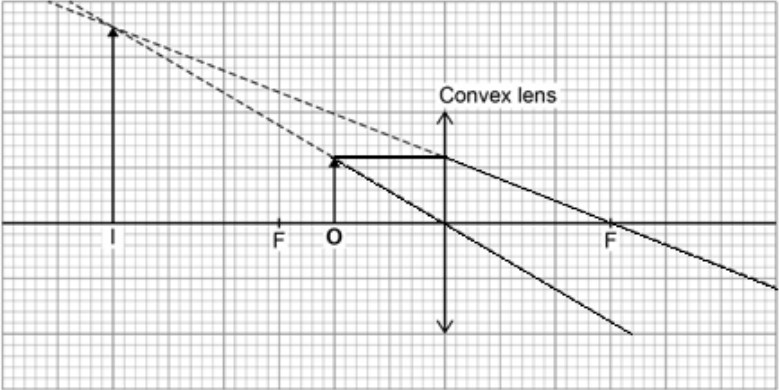
Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2	it is moving faster than any other (known) galaxy		1	AO1 3.8.3 b
	so is the furthest (known) galaxy from us		1	
		if no other mark awarded, allow 1 mark for it is moving very quickly so is very far away		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3	high energy electromagnetic radiation (present shortly after the Big Bang)	allow electromagnetic radiation that is observable in all directions	1	AO1 3.8.3 c
	whose wavelength has increased		1	
	as the Universe expanded	dependent on at least one other mark being awarded	1	
	so the Universe must have been smaller in the past		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	new evidence		1	AO3 3.8.3 d
	that cannot be explained using the Big Bang theory		1	

Total			10	
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Question 6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	a ray parallel to the principal axis and passing through the principal focus to the right of the lens a ray through the centre of the lens image drawn where the extended rays cross and resting on the principal axis 	maximum 1 mark if ruler not used to draw rays allow solid or dashed lines for rays ignore arrows on rays allow a ray drawn as if from the principal focus to the left of the lens, emerging parallel to the principal axis allow image drawn within 1 small square of point where rays cross	1 1 1	AO2 3.3.6 f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	magnification = $\frac{\text{their image height}}{\text{the object height}}$ an answer consistent with their ray diagram	allow $\frac{\text{their image distance}}{\text{the object distance}}$ allow measurements within 1 small square	1 1	AO2 3.3.6 g

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	any two from: • moves closer to lens • becomes smaller • remains virtual • remains upright	ignore remains magnified	2	AO3 3.3.6 b

Total Question 6		7
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Question 7

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	any two from - to reduce the potential difference - to provide a suitable current (to charge the phone) - to ensure the phone is not damaged	allow mains potential difference is too high	2	1 × AO1 1 × AO2 3.6.2 d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.2	smaller/lighter (than a basic transformer)	allow easier to carry	1	AO1 3.6.2 g
	uses very little power when phone is not charging	allow wastes less energy	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.3	alternating current in primary coil		1	AO1 3.6.2 c
	creates a changing magnetic field (in the iron core)		1	
	so the secondary coil experiences a changing magnetic field		1	
	inducing an (alternating) potential difference across the secondary coil		1	
	so there is an (alternating) current in the lamp	allow there is an (alternating) current in the secondary coil (because there is a complete circuit)	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.4	$\frac{1100}{50} = \frac{220}{V_s}$		1	AO2 3.6.2 e f
	$V_s = \frac{50 \times 220}{1100}$		1	
	$V_s = 10 \text{ V}$		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.5	50 mA = 0.05 A	ecf from Question 07.4	1	AO2 3.6.2 e f
	$220 \times 0.05 = 10 \times I_s$	allow a correct substitution using an incorrectly / not converted value for current	1	
	$I_s = \frac{220 \times 0.05}{10}$	allow a correct rearrangement using an incorrectly / not converted value for current	1	
	$I_s = 1.1 \text{ (A)}$	allow a correct calculation using an incorrectly / not converted value for current	1	
		a correct calculation using input current and ratio of turns can score 4 marks		

Total Question 7		16
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Question 8

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.1	measure original length and extended length		1	AO4 3.1.1 f
	take measurements at eye level	allow place the ruler close to the spring (to prevent parallax error)	1	
	subtract original length from extended length		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.2	when the aeroplane is released elastic potential energy (of the spring) decreases and kinetic energy (of the aeroplane) increases		1	AO2 3.2.2 a c
	then kinetic energy decreases and gravitational potential energy increases		1	
	energy is dissipated to the surroundings	allow work is done against air resistance	1	
	kinetic energy is zero at maximum height or gravitational potential energy is maximum at maximum height		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.3	$E_e = \frac{1}{2} \times 40 \times 0.30^2$		1	AO2 3.2.1 c d
	$E_e = 1.8 \text{ J}$	the equation $E_e = \frac{1}{2} \times k \times e^2$ must have been used to score subsequent marks	1	
	$1.8 = 0.020 \times 9.8 \times h$	allow substitution of their calculated value of energy	1	
	$h = \frac{1.8}{(0.020 \times 9.8)}$	allow a correct rearrangement using an incorrectly / not converted value of mass	1	
	$h = 9.2 \text{ (m)}$	allow a correct rearrangement using their calculated value of energy allow 9.18 (m)	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.4	elastic potential energy proportional to extension ²	allow $\frac{1}{2} \times 40 \times 0.60^2$	1	AO3 3.2.1 c
	elastic potential energy increases by a factor of 4	allow the elastic energy will be 7.2 J	1	

Total			14	
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